

Unit 5 : Regression and Classification

- **Linear Regression :-**

Linear Regression is a statistical method that is used for predictive analysis. Linear regression makes predictions for continuous/real or numeric variables such as sales, salary, age, product price etc.

Linear regression algorithm shows a linear relationship between a dependent (y) and one or more independent (x) variables, hence called as linear regression.

Since linear regression shows the linear relationship, which means it finds how the value of the dependent variable is changing according to the value of the independent variable.

Types of Linear Regression :-

1. Simple linear regression :

It is the simplest form of linear regression, where we have one independent variable (X) and one dependent variable (Y).

2. Multiple linear regression :

This is an extension of simple linear regression with two or more independent variables ($X_1, X_2, \dots$) predicting one dependent variable (Y).

- **Assessing Performance of Regression :-**

Regression performance is the ability of a regression model to make predictions that are close to the true values.

To assess regression model performance, use metrics that measure the difference between actual and predicted values, such as Mean Absolute Error (**MAE**), Mean Squared Error (**MSE**), and Root Mean Squared Error (**RMSE**).

Error Measures :-

Error measures are crucial in machine learning and statistics as they help measure the difference between predicted values and actual values.

Here are some common error measures used in machine learning:

1. Mean Absolute Error (MAE):

MAE measures the average absolute difference between predicted and actual values.

Formula: $MAE = 1/n \sum |actual - predicted|$

It gives equal weight to all errors, regardless of their magnitude.

2. Mean Squared Error (MSE):

MSE measures the average squared difference between predicted and actual values.

Formula: $MSE = 1/n \sum (Actual - Predicted)^2$

It gives more weight to larger errors compared to MAE.

3. Root Mean Squared Error (RMSE):

RMSE is the square root of MSE and provides the error in the same unit as the target variable.

Formula: $RMSE = \sqrt{MSE}$

• Overfitting and Underfitting :-

Overfitting and underfitting are common issues in machine learning and are related to the performance and generalization ability of a model.

→ Overfitting :

1. Definition :

Overfitting occurs when a machine learning model learns the training data too well, capturing noise, random fluctuations, and outliers in the data, instead of just the underlying patterns and relationships.

As a result, the model performs very well on the training data but poorly on unseen or new data.

2. Characteristics :

Low training error (fits the training data closely).

High testing error (poor performance on new data).

Complex models, such as those with too many parameters or high polynomial degrees, are more prone to overfitting.

3. Causes:

Too complex a model relative to the amount of training data.

Insufficient data pre-processing, such as not removing outliers or irrelevant features.

Excessive training time or iterations, allowing the model to memorize the training data.

4. Mitigation:

Use simpler models or regularization L_1 or L_2 techniques (for example, regularization).

Increase the amount of training data.

Perform feature selection or engineering to reduce dimensionality.

Cross-validation to tune hyperparameters and assess model performance on multiple subsets of the data.

→ Underfitting:

1. Definition:

Underfitting occurs when a machine learning model is too simple to capture the underlying patterns in the data.

It fails to learn the training data sufficiently, resulting in poor performance on both the training data and new data.

2. Characteristics:

High training error (fails to fit the training data).

High testing error (poor performance on new data).

Simple models that cannot capture the data's complexity are prone to underfitting.

3. Causes:

Using a model that is too simple for the problem at hand.

Insufficient feature engineering or selection.

Insufficient training time or iterations, preventing the model from learning complex patterns.

4. Mitigation :

Use more complex models or increase model capacity.

Perform feature engineering to expose more relevant information in the data.

Ensure sufficient training time or iterations.

Collect more data if possible, to better represent the problem's complexity.

• **Catalyst for overfitting :-**

Overfitting occurs when a machine learning model memorizes the training data instead of learning the underlying patterns.

This results in high accuracy on the training set but poor performance on unseen data.

Key Factors That Cause Overfitting:

1. Excessively Complex Models -

Using a model that is too complex for the given dataset, such as deep neural networks with too many layers or high-degree polynomial regression, leads to overfitting.

2. Small Training Dataset -

If the dataset is too small, the model memorizes specific examples instead of generalizing patterns.

3. Too Many Features -

When the number of input features is too high compared to the number of observations, the model has too much flexibility.

4. Excessive Training Time -

If a model is trained for too many iterations, it can start memorizing noise in the data rather than general patterns.

5. Noisy Data or Outliers -

If the training data contains too much noise or outliers, a highly flexible model may try to learn these random variations.

Methods to Avoid Overfitting :

1. Use More Training Data: A larger dataset helps the model generalize better.
2. Reduce Model Complexity: Simplify the model by reducing the number of parameters or using a smaller neural network.
3. Early Stopping: Stop training when the validation loss stops decreasing.
4. Cross-Validation: Use K-Fold Cross-Validation to assess model performance on multiple data subsets.

• Catalyst for Underfitting :-

Underfitting occurs when a machine learning model is too simple to capture the underlying structure of the data, leading to both high training error and high test error.

Key Factors That Cause Underfitting :

1. Over-Simplified Model -

Choosing a model that is too simple prevents it from capturing the complexity of the dataset.

This results in a poor fit and inaccurate predictions.

2. Insufficient Training Time -

In neural networks, stopping training too early can prevent the model from learning the full pattern.

3. Too Few Features -

When important features are missing, the model has limited information to make accurate predictions.

4. Inadequate Data Preprocessing -

Poor feature scaling, missing value imputation, or failure to encode categorical variables properly can lead to a poorly trained model.

5. Too Few Training Examples -

If there are too few training examples, the model may not have enough data to learn from.

Methods to Avoid Underfitting :

1. Use a More Complex Model: Consider using polynomial regression, decision trees, or deep neural networks for non-linear problems.
 2. Train for More Epochs: In deep learning, increasing the number of training epochs can help the model learn better.
 3. Feature Engineering: Add meaningful features that provide more relevant information.
 4. Increase Training Data: More data helps the model capture the full complexity of the problem.
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• Multiple Linear Regression :-

Multiple Linear Regression is a statistical technique used to model the relationship between one dependent variable (continuous) and two or more independent variables (predictors).

It's basically an extension of simple linear regression, which uses only one independent variable.

Difference from Simple Linear Regression: Simple uses one predictor; MLR uses two or more predictors.

Applications:

Economics, marketing, real estate, health sciences — anywhere multiple factors influence one outcome.

Purpose :

To predict the value of a dependent variable based on several independent variables.

To understand the impact of each predictor while controlling for others.

The MLR Equation :

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n + \epsilon$$

In Words,

Y = the thing you're trying to predict (dependent variable).

$X_1, X_2 \dots X_n$ = the different factors (independent variables) you think affect Y .

β_0 (intercept) = the starting value of Y when all X 's are zero.

$\beta_1, \beta_2 \dots \beta_n$ (coefficients) = how much Y changes when each X changes by one unit (while keeping the other X 's fixed).

ε (error) = the difference between your prediction and the actual value (random noise).

• Metrics for Regression :-

When evaluating the performance of regression models, several metrics are commonly used to assess how well the model fits the data and makes predictions.

Here are some of the most commonly used regression metrics :

I. Mean Squared Error (MSE)

Mean squared error (MSE) measures the amount of error in statistical models.

It assesses the average squared difference between the observed and predicted values.

As model error increases, its value increases.

The mean squared error is also known as the mean squared deviation (MSD).

Pros:

Penalizes larger errors more heavily (good if large errors are especially bad).

Cons:

Sensitive to outliers.

 Formula

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Where:

- n = number of observations
- y_i = actual value
- $\hat{y}_i$ = predicted value

In Simple Words

Take each prediction, subtract the actual value, square the result, then average all those squares.

II. Root mean squared error (RMSE)

Root Mean Squared Error is the square root of the average of the squared errors between predicted and actual values.

It's just the square root of MSE, so it brings the error back into the same units as your target variable.

Mathematically, it is the standard deviation of the residuals. Residuals represent the distance between the regression line and the data points. RMSE is always non-negative (zero means perfect prediction).

Pros:

Easy to interpret because units match the target variable.

Cons:

Like MSE, sensitive to outliers.

 Formula

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

Where:

- n - number of observations
- y_i - actual value
- $\hat{y}_i$ - predicted value

In Simple Words

Take each prediction error, square it, average all those squares (that's MSE).

Then take the square root → that's RMSE.

III. Mean absolute error (MAE)

Mean Absolute Error (MAE) is the average of the absolute differences between the actual values and the predicted values.

It tells you, on average, how far your model's predictions are from the real values — without squaring the errors.

Pros:

Less sensitive to outliers than MSE/RMSE.

Cons:

Doesn't penalize large errors as strongly as MSE.

2 Formula

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

Where:

- n - number of observations
- y_i - actual value
- $\hat{y}_i$ - predicted value

In Simple Words

Take each prediction, subtract the actual value → error.

Take the absolute value (ignore minus sign).

Average all those absolute errors.

That's your MAE.

• Logistic Regression :-

Logistic Regression is a supervised machine learning algorithm used for classification problems.

Unlike linear regression which predicts continuous values it predicts the probability that an input belongs to a specific class.

It is used for binary classification where the output can be one of two possible categories such as Yes/No, True/False or 0/1.

It is mainly used to classify data into classes like yes/no, spam/not spam, disease/no disease, etc.

It uses sigmoid function to convert inputs into a probability value between 0 and 1.

The sigmoid function is an important part of logistic regression which is used to convert the raw output of the model into a probability value between 0 and 1.

Applications :

Medical diagnosis (disease vs no disease).

Credit scoring (default vs no default).

Spam detection (Spam vs no spam).

Marketing (purchase vs not purchase).

Goal of Logistic Regression :

1. Predict the probability that a given input belongs to a certain class (usually Class 1).

2. Use that probability to make a classification decision (yes/no, success/failure, spam/not spam).

- **Binary and Multiclass Classification :-**

Binary classification is the simplest form of classification, where the target variable has only two possible classes or outcomes.

Examples -

Email: Spam (1) or Not Spam (0)

Loan: Default (1) or No Default (0)

Medical test: Disease (1) or No Disease (0)

Multiclass classification is when the outcome variable has three or more categories.

Example -

Fruit type: Apple (0) / Banana (1) / Mango (2) / Orange (3)

Animal category: Cat (0) / Dog (1) / Bird (2) / Horse (3)

Weather condition: Sunny (0) / Rainy (1) / Cloudy (2) / Snowy (3)

Binary classification is the simplest and most common starting point.

Multiclass classification expands logistic regression's usefulness to real-world problems where you have many possible outcomes.

The key difference is the number of possible outcomes: two for binary and more than two for multiclass classification.

Assessing Classification Performance :

Classification is about predicting a label and then identifying which category an object belongs to based on different parameters.

In order to measure how well our classification model is doing at making these predictions, we use classification metrics.

1. Confusion Matrix :

A confusion matrix compares the actual classes vs. the predicted classes.

2. ROC Curve & AUC :

ROC Curve: Plots True Positive Rate (Recall) vs. False Positive Rate at different thresholds.

AUC (Area Under the Curve): A single number showing model's ability to discriminate between classes.

3. F1 Score :

Harmonic mean of precision and recall (balances both)

Useful if you want a single metric combining precision and recall.

4. Accuracy :

Proportion of correct predictions (overall):

Good for balanced datasets.

Example: 90 correct out of 100 = 90% accuracy.

5. Precision :

Out of all predicted positives, how many are actually positive?

High precision = few false positives.

6. Recall :

Out of all actual positives, how many did we correctly predict?

High recall = few false negatives.

Handling more than two classes :-

Logistic regression, by default, is limited to two-class classification problems. Some extensions like one-vs-rest can allow logistic regression to be used for multi-class classification problems, although they require that the classification problem first be transformed into multiple binary classification problems.

- **Multiclass Classification :**

Multiclass classification is a type of supervised machine learning problem where an algorithm learns to categorize each input into one of three or more possible classes (labels) based on its features.

Input = data with multiple features.

Output = one label chosen from several categories.

Binomial (or Binary) Logistic Regression is a statistical / machine learning technique used to predict the probability of one of two possible outcomes (yes/no, 0/1) based on one or more input variables.

Multinomial Logistic Regression (also called Softmax Regression) is an extension of logistic regression used when the dependent variable (output) has three or more categories (not just two).

Binary Classifiers for multi-class classification :

1. One vs One :-

One-vs-One (OvO) is a strategy for handling multiclass classification by training a separate binary classifier for every possible pair of classes, then combining the results to decide the final class. It's also called pairwise classification.

Example:

You have 3 classes of fruits: Apple (A), Banana (B), Orange (C).

Training (building models)

OvO will create one binary classifier for each pair:

Model 1: Apple vs Banana

Model 2: Apple vs Orange

Model 3: Banana vs Orange

So total = 3 classifiers for 3 classes.



Prediction (new fruit comes in)

A new fruit is shown to the system. Each model gives its

vote:

Model	Which class it picks?
Apple vs Banana	Apple
Apple vs Orange	Orange
Banana vs Orange	Orange

Count votes:

Apple = 1 vote

Banana = 0 votes

Orange = 2 votes

Final Prediction = Orange (highest votes)

2. One vs Rest :-

One-vs-Rest (OvR), also called One-vs-All, is a strategy for multiclass classification where one classifier is trained for each class, treating that class as positive and all other classes combined as negative.

Example:

You have 3 classes of fruits: Apple (A), Banana (B), Orange (C).



Training (building models)

OvR creates one binary classifier for each class:

Classifier	Positive Class	Negative Class
Model 1	Apple	Banana + Orange
Model 2	Banana	Apple + Orange
Model 3	Orange	Apple + Banana



Prediction (new fruit comes in)

A new fruit is shown to the system. Each model predicts a score/probability for its class:

Model	Probability
Apple vs Rest	0.6
Banana vs Rest	0.3
Orange vs Rest	0.1



Final Prediction = Apple (highest probability)

- **Metrics for Classification :-**

Classification metrics are used to evaluate the performance of a classification model, i.e., how well it predicts categories or classes.

i. Confusion Matrix :

The confusion matrix is the foundation of classification metrics. It compares predicted labels vs true labels.

Explanation of terms:

True Positive (TP): Correctly predicted positive samples.

True Negative (TN): Correctly predicted negative samples.

False Positive (FP): Predicted positive but actually negative. Also called Type I Error.

False Negative (FN): Predicted negative but actually positive. Also called Type II Error.

This matrix is essential because all other metrics are derived from it.

2. Accuracy :

The proportion of total correct predictions.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Simple but can be misleading if classes are imbalanced.

3. Precision :

How many predicted positives are actually positive.

$$\text{Precision} = \frac{TP}{TP + FP}$$

High precision → fewer false positives.

4. Recall (Sensitivity / True Positive Rate) :

How many actual positives are correctly predicted.

$$\text{Recall} = \frac{TP}{TP + FN}$$

High recall → fewer false negatives.

5. F1-Score :

Harmonic mean of precision and recall. Balances both.

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Useful when classes are imbalanced.

6. ROC-AUC (Receiver Operating Characteristic – Area Under Curve) :

Plots True Positive Rate vs False Positive Rate.

AUC score measures model's ability to distinguish between classes (0.5 → random, 1 → perfect).